EXPERIMENT NO: 07

Experiment Title: To perform DOM traversal and manipulation, and to develop a simple To-Do List application with Add, Edit, and Delete functionalities using HTML, CSS, and JavaScript.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory:

The Document Object Model (DOM) represents an HTML document as a tree of objects. JavaScript can access and modify these objects using DOM methods such as getElementById(), querySelector(), createElement(), and appendChild(). DOM manipulation allows web pages to be updated dynamically without reloading the page.

In a To-Do List application, JavaScript is used to create new task elements, modify existing tasks, and remove tasks from the DOM. The main operations performed are:

- Add: Creates and displays a new task.
- Edit: Modifies the text or properties of an existing task.
- Delete: Removes a task element from the DOM.
- DOM Traversal: Accesses and navigates through HTML elements to perform these updates.

Experiment Program Code

Code 1: Query_Selector.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>querySelector() Example</title>

    <style>
        body{
            font-family: Arial, Helvetica, sans-serif;
            text-align: centre;
            margin-top: 50px;
        }

        .container{
            width : 500px;
            margin : auto;
            padding : 30px;
            border : 2px solid black;
            border-radius : 10px;
        }
```

```
        button{
            padding : 10px 20px;
            font-size: 16px;
            cursor : pointer ;
        }
    </style>
</head>
<body>
    <div class = "container">
        <h2 id = "heading">Javascript DOM Manipulation</h2>
        <p class = "message">
            This is the original message
        </p>
        <button onclick="changeContent()">
            Change Button
        </button>
    </div>

    <script>
        function changeContent(){
            let heading = document.querySelector('#heading');
            let message = document.querySelector('#message');

            heading.textContent = "Welcome to DOM Manipulation";

            message.textContent = "Content changed using querySelector()";

            heading.style.color = "blue";
            mesage.style.fontsize = "20px";
        }
    </script>
</body>
</html>
```

4. Output



5. Case Study Title

Design a To-Do List application using HTML, CSS and Javascript.

6. Case Study Program Code

Casestudy_06_Shweta_Kushwaha_1001.html

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Smart To-Do List</title>

    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: Arial, sans-serif;
        }

        body {
            min-height: 100vh;
            background: #f6f1e8;
            color: #3f372f;
            padding: 35px 20px;
        }

        .container {
            max-width: 900px;
            margin: auto;
        }

        /* HEADER */

        header {
            text-align: center;
            margin-bottom: 30px;
        }

        header .icon {
            font-size: 45px;
            margin-bottom: 8px;
        }

        header h1 {
            font-size: 36px;
            color: #3d342d;
            margin-bottom: 8px;
        }

        header p {
            color: #8a7d70;
            font-size: 15px;
        }

        /* MAIN CARD */

        .main-card {
            background: white;
            border: 1px solid #e6ddd2;
            border-radius: 22px;
            padding: 28px;
            box-shadow: 0 10px 30px rgba(87, 69, 49, 0.08);
        }

        /* INPUT AREA */

        .input-area {
            display: flex;
            gap: 10px;
            margin-bottom: 25px;
        }

        #taskInput {
            flex: 1;
            padding: 14px 16px;
            border: 1px solid #ded3c6;
            border-radius: 11px;
            background: #fcfaf7;
            color: #3f372f;
            font-size: 15px;
            outline: none;
        }

        #taskInput:focus {
```

```
            border-color: #b58664;
            box-shadow: 0 0 0 3px rgba(181, 134, 100, 0.12);
        }

        #addBtn {
            padding: 14px 22px;
            border: none;
            border-radius: 11px;
            background: #9a6b4d;
            color: white;
            font-size: 15px;
            font-weight: bold;
            cursor: pointer;
        }

        #addBtn:hover {
            background: #81573e;
        }

        /* STATISTICS */

        .stats {
            display: grid;
            grid-template-columns: repeat(3, 1fr);
            gap: 12px;
            margin-bottom: 25px;
        }

        .stat-box {
            background: #faf7f2;
            border: 1px solid #e8dfd5;
            border-radius: 14px;
            padding: 16px;
            text-align: center;
        }

        .stat-number {
            display: block;
            font-size: 25px;
            font-weight: bold;
            color: #9a6b4d;
            margin-bottom: 5px;
        }

        .stat-label {
            color: #8a7d70;
            font-size: 13px;
        }

        /* SECTION HEADER */

        .list-header {
            display: flex;
            justify-content: space-between;
            align-items: center;
            margin-bottom: 15px;
        }

        .list-header h2 {
            font-size: 21px;
        }

        #clearBtn {
            border: none;
            background: #f4e9df;
            color: #8a6045;
            padding: 8px 13px;
            border-radius: 8px;
            cursor: pointer;
            font-size: 13px;
        }

        #clearBtn:hover {
            background: #ead9c9;
        }

        /* TASK LIST */
```

```
#taskList {
    list-style: none;
}

.task-item {
    display: flex;
    align-items: center;
    gap: 12px;
    padding: 14px;
    margin-bottom: 10px;
    background: #fcfaf7;
    border: 1px solid #e7ddd2;
    border-radius: 12px;
    transition: 0.2s;
}

.task-item:hover {
    border-color: #d2bda9;
    transform: translateY(-1px);
}

.check-btn {
    width: 22px;
    height: 22px;
    border-radius: 50%;
    border: 2px solid #b89576;
    background: white;
    cursor: pointer;
    flex-shrink: 0;
}

.task-text {
    flex: 1;
    font-size: 15px;
    color: #51463c;
    word-break: break-word;
}

.completed .task-text {
    text-decoration: line-through;
    color: #a59b91;
}

.completed .check-btn {
    background: #9a6b4d;
    border-color: #9a6b4d;
}

/* ACTION BUTTONS */

.actions {
    display: flex;
    gap: 6px;
}

.edit-btn,
.delete-btn {
    border: none;
    padding: 7px 10px;
    border-radius: 7px;
    cursor: pointer;
    font-size: 12px;
}

.edit-btn {
    background: #eee6dc;
    color: #755b46;
}

.delete-btn {
    background: #f5e4df;
    color: #a05d51;
}

.edit-btn:hover {
    background: #e3d6c7;
```

```
        }

        .delete-btn:hover {
            background: #ead0c9;
        }

        /* EMPTY MESSAGE */

        .empty {
            text-align: center;
            padding: 45px 20px;
            color: #a3978c;
        }

        .empty-icon {
            font-size: 40px;
            margin-bottom: 10px;
        }

        .empty p {
            font-size: 14px;
        }

        /* FOOTER */

        footer {
            text-align: center;
            margin-top: 20px;
            color: #a3978c;
            font-size: 12px;
        }

        /* RESPONSIVE */

        @media (max-width: 600px) {

            body {
                padding: 20px 12px;
            }

            .main-card {
                padding: 20px;
            }

            header h1 {
                font-size: 29px;
            }

            .input-area {
                flex-direction: column;
            }

            #addBtn {
                width: 100%;
            }

            .stats {
                grid-template-columns: 1fr;
            }

            .task-item {
                align-items: flex-start;
            }

            .actions {
                flex-direction: column;
            }
        }
    </style>
</head>

<body>

<div class="container">

    <!-- HEADER -->
```

```
<header>
    <div class="icon">✓</div>
    <h1>Smart To-Do List</h1>
    <p>Manage your daily tasks using JavaScript DOM Manipulation</p>
</header>

<!-- MAIN -->

<div class="main-card">

    <!-- INPUT -->

    <div class="input-area">

        <input
            type="text"
            id="taskInput"
            placeholder="What do you need to do?"
        >

        <button id="addBtn" onclick="addTask()">
            + Add Task
        </button>

    </div>

    <!-- STATISTICS -->

    <div class="stats">

        <div class="stat-box">
            <span class="stat-number" id="totalTasks">0</span>
            <span class="stat-label">Total Tasks</span>
        </div>

        <div class="stat-box">
            <span class="stat-number" id="completedTasks">0</span>
            <span class="stat-label">Completed</span>
        </div>

        <div class="stat-box">
            <span class="stat-number" id="pendingTasks">0</span>
            <span class="stat-label">Pending</span>
        </div>

    </div>

    <!-- TASK HEADER -->

    <div class="list-header">

        <h2>My Tasks</h2>

        <button id="clearBtn" onclick="clearTasks()">
            Clear All
        </button>

    </div>

    <!-- TASK LIST -->

    <ul id="taskList">

        <div class="empty" id="emptyMessage">

            <div class="empty-icon">📝</div>

            <p>
                No tasks yet. Add your first task!
            </p>

        </div>

    </ul>

</div>
```

```
    <footer>
        JavaScript Lab Project • DOM Traversal • DOM Manipulation • Events
    </footer>

</div>

<script>

    /* --------------------------------
       ADD TASK
    -------------------------------- */

    function addTask() {

        let input =
            document.getElementById("taskInput");

        let taskText =
            input.value.trim();

        // Check for empty task
        if (taskText === "") {

            alert("Please enter a task.");

            return;
        }

        // Create new list item
        let li =
            document.createElement("li");

        li.className = "task-item";

        // Create check button
        let checkButton =
            document.createElement("button");

        checkButton.className = "check-btn";

        checkButton.onclick =
            function() {

                toggleTask(li);

            };

        // Create task text
        let span =
            document.createElement("span");

        span.className = "task-text";

        span.textContent = taskText;

        // Create action container
        let actions =
            document.createElement("div");

        actions.className = "actions";

        // Edit button
        let editButton =
            document.createElement("button");

        editButton.className = "edit-btn";

        editButton.textContent = "Edit";

        editButton.onclick =
            function() {

                editTask(span);

            };

        // Delete button
```

```
        let deleteButton =
            document.createElement("button");

        deleteButton.className = "delete-btn";

        deleteButton.textContent = "Delete";

        deleteButton.onclick =
            function() {

                deleteTask(li);

            };

        // Add buttons to actions
        actions.appendChild(editButton);

        actions.appendChild(deleteButton);

        // Add everything to list item
        li.appendChild(checkButton);

        li.appendChild(span);

        li.appendChild(actions);

        // Add list item to task list
        document
            .getElementById("taskList")
            .appendChild(li);

        // Clear input
        input.value = "";

        input.focus();

        // Hide empty message
        document
            .getElementById("emptyMessage")
            .style.display = "none";

        updateStatistics();
    }

    /* --------------------------------
       COMPLETE / UNCOMPLETE TASK
    -------------------------------- */

    function toggleTask(task) {

        task.classList.toggle("completed");

        updateStatistics();
    }

    /* --------------------------------
       EDIT TASK
    -------------------------------- */

    function editTask(taskText) {

        let newText =
            prompt(
                "Edit your task:",
                taskText.textContent
            );

        if (newText !== null) {

            newText = newText.trim();

            if (newText !== "") {

                taskText.textContent =
                    newText;

            } else {
```

```
            alert("Task cannot be empty.");

        }

    }

}

/* --------------------------------
   DELETE TASK
-------------------------------- */

function deleteTask(task) {

    task.remove();

    updateEmptyMessage();

    updateStatistics();
}

/* --------------------------------
   CLEAR ALL TASKS
-------------------------------- */

function clearTasks() {

    let taskList =
        document.getElementById("taskList");

    // DOM traversal
    let tasks =
        taskList.querySelectorAll(".task-item");

    if (tasks.length === 0) {

        return;

    }

    if (confirm("Delete all tasks?")) {

        tasks.forEach(function(task) {

            task.remove();

        });

        updateEmptyMessage();

        updateStatistics();

    }

}

/* --------------------------------
   UPDATE EMPTY MESSAGE
-------------------------------- */

function updateEmptyMessage() {

    let taskList =
        document.getElementById("taskList");

    let tasks =
        taskList.querySelectorAll(".task-item");

    let emptyMessage =
        document.getElementById("emptyMessage");

    if (tasks.length === 0) {

        emptyMessage.style.display =
            "block";
```

```
        } else {

            emptyMessage.style.display =
                "none";

        }

    }

    /* --------------------------------
       UPDATE STATISTICS
    -------------------------------- */

    function updateStatistics() {

        let taskList =
            document.getElementById("taskList");

        // DOM traversal
        let tasks =
            taskList.querySelectorAll(".task-item");

        let completed =
            taskList.querySelectorAll(
                ".task-item.completed"
            );

        let total =
            tasks.length;

        let completedCount =
            completed.length;

        let pending =
            total - completedCount;

        // Update DOM
        document.getElementById("totalTasks")
            .textContent = total;

        document.getElementById("completedTasks")
            .textContent = completedCount;

        document.getElementById("pendingTasks")
            .textContent = pending;

    }

    /* --------------------------------
       ENTER KEY SUPPORT
    -------------------------------- */

    document
        .getElementById("taskInput")
        .addEventListener(
            "keypress",
            function(event) {

                if (event.key === "Enter") {

                    addTask();

                }

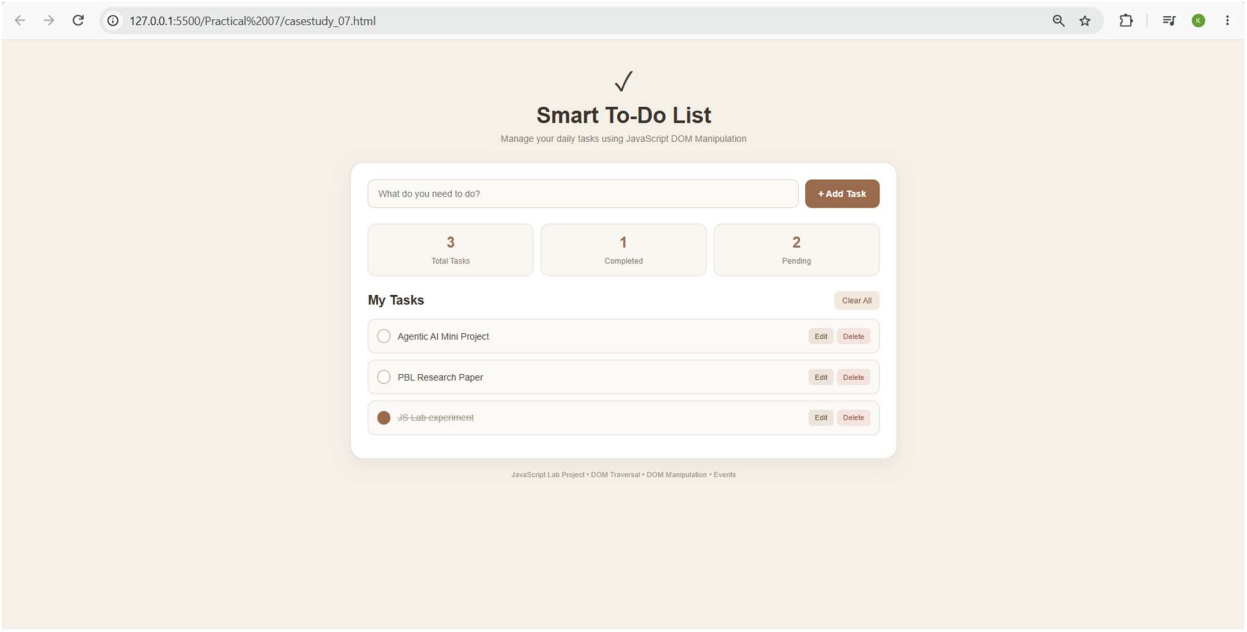
            }
        );

</script>

</body>
</html>
```

7. Output of the case study :

Name : Shweta Kushwaha PRN : 24070521001 File path :
http://127.0.0.1:5500/Practical%2002/Casestudy_07_Shweta_Kushwaha_1001.html



8. Conclusion

The To-Do List application was successfully developed using HTML, CSS, and JavaScript. DOM traversal and manipulation were performed to dynamically add, edit, and delete tasks from the webpage.